$1 + 2 + 3 + 4 + - + n = \frac{n \cdot (n+1)}{2}$.

Base Case: for $n=1$, verify that the formula is true.

Inductive Step:

Arbitrary natural number k_0.

$1 + 2 + - + k_0 = \frac{k_0 \cdot (k_0+1)}{2}$.

$F(k_0) \rightarrow F(k_0+1)$

$$\frac{P(n=1) \wedge \forall k \; P(k) \rightarrow P(k+1)}{\forall n. \; P(n) \equiv T.}$$

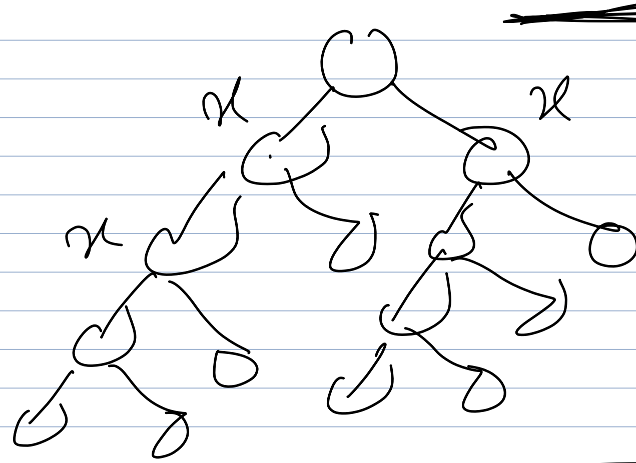
Strong Mathematical Induction:

$n=1$ $\quad$ $n = k_0$ | $n := k_0+1$

To prove: An RB-tree with n internal nodes has height at max. $2 \cdot \lg(n+1)$

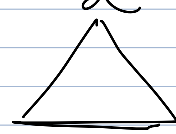
Any sub-tree rooted at node x has at the least $2^{bh(x)} - 1$ internal nodes.

$bh(x)$



$\boxed{bh(x)}$

x



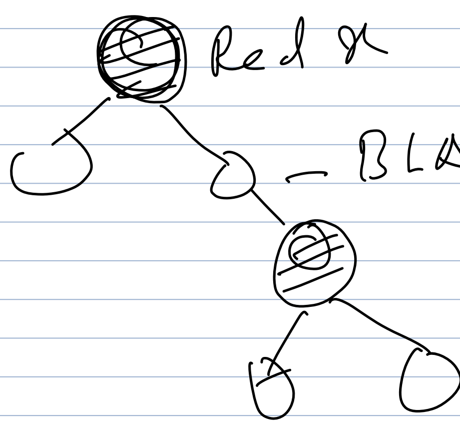
$2^{bh(x)} - 1$

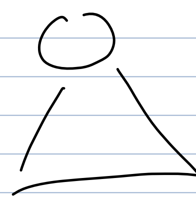
internal nodes.

x = Total Tree Root node.

Min $\boxed{2^{bh(x)} - 1}$

$h/2$ black nodes





$\boxed{2^{bh(x)} - 1}$

Min. Black Height $(x) = \frac{h}{2}$.

Min Number of internal nodes

$N = 2^{bh(x)} - 1$

$$N = 2^{h/2} - 1$$

$$N+1 = 2^{h/2}.$$

$$2^{h/2} = N+1 \qquad \log_2 2^n = n$$

$$\log_2 2^{h/2} = \log_2 (N+1)$$

$$h/2 = \log_2 (N+1)$$

$$h = 2 \cdot \log_2 (N+1)$$

$$N \geq 2^{h/2} - 1$$

$$N+1 \geq 2^{h/2}.$$

$$2^{h/2} \leq N+1$$

$$h/2 \leq \log_2 (N+1)$$

$$\boxed{h \leq 2 \cdot \log_2 (N+1)}$$

$$N \geq 2^{bh(x)} - 1$$

$$N \geq 2^{h/2} - 1$$

$$2^{h/2} - 1 \leq N. \qquad \log_x y$$

$$2^{h/2} \leq N+1$$

$$x^{?} = y$$

$$\log_2 2^{h/2} \leq \log_2 (N+1)$$

$$2^{?} = 2^{h/2}$$

$$\frac{h}{2} \leq \log_2 (N+1)$$

$$h \leq \log_2 (N+1)$$